

EFFECTS OF VARIOUS WARM-UP PROTOCOL ON SPECIAL JUDO FITNESS TEST PERFORMANCE

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ABSTRACT

Lum, D. Effects of various warm-up protocol on special judo fitness test performance. *J Strength Cond Res* 33(2): 459–465, 2019—The purposed of this study was to compare the effects of postactivation potentiation (PAP) on Special Judo Fitness Test (SJFT) performance using explosive exercises that activates upper and lower limbs muscles. Eleven male judo athletes (mean $\pm$ SD, age, 16–29 years; height, 170 ± 7 cm; body mass, 73 ± 16 kg) attended 4 separate sessions. The first session was used to familiarize the subjects to the experimental procedure, the SJFT, the high pull test (HPT), and the 2 explosive exercises including resistance band pull and standing broad jump. Subsequently, subjects were randomly assigned in a counterbalanced manner to either perform the upper and lower body (ULB) PAP, lower body (LB) PAP or usual competition (CON) warm-up routine before performing the HPT and SJFT. The following variables were quantified: throws performed during series A, B, and C; total number of throws; heart rate immediately and 1 minute after the test; test index; peak power; and RPE after warm-up. During series 1, number of throws performed in LB and ULB were significantly greater than CON ($p \leq 0.05$). Only ULB resulted in significantly greater number of total throws ($p < 0.01$) and higher peak power ($p < 0.01$) than CON. The RPE for both LB and ULB were significantly lower than CON ($p < 0.01$). Peak power was moderately correlated to total number of throws performed ($r = 0.4$, $p \leq 0.05$). This study suggest that performing ULB before SJFT can result in improved performance and peak power.

KEY WORDS postactivation potentiation, variable resistance, combat sports, plyometrics

INTRODUCTION

Warming up before athletic event is a common practice. It has been suggested that sporting performance can be enhanced by performing a warm-up routine via various mechanisms including decreased resistance of muscle and joints; increased

rate of metabolic reaction; higher release of oxygen from both myoglobin and hemoglobin; increased rate of nerve conduction; increased blood flow to working muscles; post-activation potentiation (PAP); and improved psychological preparedness (3).

One of the mechanisms that leads to improvement in performance after warm-up is known as postactivation potentiation, an increase in low frequency tetanic force and muscle twitch after activities that induce high intensity muscle contraction (19). It was proposed that PAP is a result of either an increase in the phosphorylation of myosin regulatory light chains (13) or an increase in recruitment of fast twitch motor units (12). To induce PAP, practitioners and researchers would require athletes to perform a high intensity resistance exercise that cause either dynamic or isometric contraction before the main sports activity. Studies have shown that PAP is able to improve plyometric jumps performance (11,18), sprint times (21,25), and sport specific activities (1,5,6,14–16,26). It has been suggested that performing multiple sets of resistance exercises at 60–84% of 1 repetition max (1RM) is superior to just doing 1 set at $>85\%$ of 1RM for inducing PAP (20,24), some studies have also shown beneficial potentiating effects of plyometrics and variable resistance exercises (2,16,17). It has been suggested that a 5–10 minutes of recovery period between the resistance exercise and sports activity is required to allow fatigue to dissipate (20,24).

Despite the supporting evidence on the positive effect of PAP on exercise performance, little is known about the effect of PAP on judo performance. To the knowledge of the author, only the study by Miarka et al. (16) has been carried out to investigate the effect of PAP on judo performance. In this study, 8 judo athletes were recruited to compare the effects of plyometric exercise, heavy resistance exercise, and mix plyometric and heavy resistance exercise as PAP stimulus, on performance of Special Judo Fitness Test (SJFT). The study showed that the plyometric stimulus resulted in significantly higher number of throws executed during the first series of the test, and the mix plyometric and heavy resistance stimulus resulted in a better index score, as compared to the control condition. Despite the positive findings, caution is advised when interpreting the results because the control condition did not perform any warm-up protocol before the SJFT. This might have affected the athletes' performance. In addition, all PAP conditions

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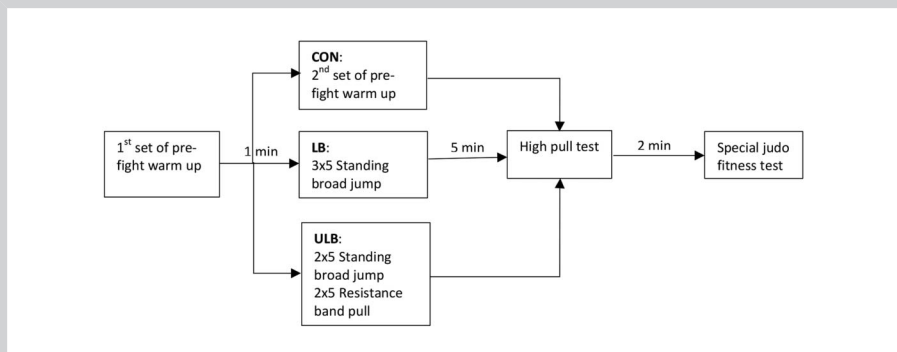


Figure 1. Timeline for the testing sessions.

used exercises that involved only the lower limbs and did not involve activation of upper limb muscles which were required in a judo throw action (10). Furthermore, it might not be possible for a judo athlete to perform heavy resistance exercise before competition because of equipment constraints. Finally, the study did not investigate the effectiveness of PAP stimulus in increasing muscular power.

Therefore, the aim of this study is to investigate the effects of PAP on SJFT performance and peak power output using explosive exercises that involved activating both upper and lower limbs muscles, and are possible to be performed at competition venues. It is hypothesized that the addition of a PAP stimulus will result in increased muscular power and superior SJFT performance as compared to a usual competition warm-up protocol. It is also hypothesized that the simultaneous inclusion of an upper body PAP stimulus will result in superior performance as compared to LB PAP stimulus alone.

METHODS

Experimental Approach to the Problem

This study used a randomized counterbalance approach. Subjects attended 4 separate sessions. The first session was

used to familiarize the subjects to the experimental procedure, the SJFT, the high pull test (HPT) and the exercises used to induce PAP. During the subsequent 3 sessions, subjects were randomly assigned in a counter-balanced manner to either perform the upper and lower body (ULB) PAP, lower body (LB) PAP or usual competition (CON) warm-up routine before performing the HPT and SJFT.

Subjects

Eleven male judo athletes (age, 16–29 years; height, 170 ± 7 cm; body mass, 73 ± 16 kg) were recruited for this study. Athletes were trained in judo for more than 2 years. Eight of the athletes have been actively competing in international competitions whereas the other 3 were ranked top 3 in the national junior tournament. All athletes were proficient at performing the ippon-seoi-nage throwing technique and have been participating in resistance training twice a week for at least 1 year. Study was conducted 1 week after the national opens tournament. Before participation, all subjects were briefed on the requirements and risks involved with the study. Subjects signed a written informed consent before the initial testing session. Parental consent was sought for subjects who were below 18 years of age. The study commenced after obtaining clearance from the Institutional Review Board at Singapore Sports Institute.

Protocols and Tests

Familiarization. Subjects reported to the judo dojo at the same time of day for all experimental sessions. During the first session, subjects were briefed on the testing methods and procedures along with their body stature and mass being

TABLE 1. Warm-up protocols.

Exercise	Volume		
	CON	LB	ULB
Jogging	5 min	5 min	5 min
Stretching	5 min	5 min	5 min
Break fall drills (Front, back, side, and rolling)	10 repetitions each	5 repetitions each	5 repetitions each
Uchikomi (throwing drills)	2 × 10 repetition	1 × 10 repetitions	1 × 10 repetitions
Rapid Uchikomi	2 × 10 repetitions	1 × 10 repetitions	1 × 10 repetitions
Komikata (fight for grip)	2 × 1 min	1 × 1 min	1 × 1 min
Nagekomi (practice throw)	2 × 10 repetitions	1 × 10 repetitions	1 × 10 repetitions
Standing broad jump		3 × 5 repetitions	2 × 5 repetitions
Resistance band pull			2 × 5 repetitions



Figure 2. Resistance band pull exercise.

measured. Subsequently, subjects performed a 10 minutes warm-up including 3 minutes of self-paced jogging, dynamic stretches for all major muscle groups and 10 repetitions of throwing drills for the ippon-seoi-nage throw technique, followed by 3 repetitions of the HPT after a 5 minutes recovery period. Subjects then performed the SJFT after 2 minutes of passive recovery. Subsequently, subjects were given 15 minutes of recovery before performing the resistance band pull to determine the resistance used during the ULB (Figure 1). Subjects performed 3 repetitions of the resistance band pull that mimicked the movement of the initial pulling phase of a judo throw at maximal effort. This movement initiates with subject extending their front knee and rotating their hips while simultaneously pulling the resistance band upward and backward with slightly flexed elbows. Subjects attempted the exercise with resistance bands (Gym In The Pocket, resistance band) of increasing resistance until they reached the resistance whereby they were unable to go through the full range of motion for the exercise. The resistance bands were attached to a rigid metal bar that was attached to the wall and 1 m above the ground. During each attempt, subjects were instructed to perform the pull as powerfully as possible. They were given 3 minutes of rest after each resistance level. It was decided that the band with the highest resistance that the participant was able to pull through the full range of motion with was used for the ULB, as there was no study done to provide a recommendation as to what resistance level induces optimal PAP effect. This exercise was selected as it was an exercise that subjects performed regularly during judo training. The subsequent 3 sessions were carried out 72 hours apart thereafter.

Warm-Up Protocols. During the subsequent 3 testing sessions, subjects reported to the judo dojo after having refrained from caffeine, alcohol and strenuous exercise for 24 hours. Subjects were randomly assigned in a counterbalance order to either perform the ULB, LB, or CON warm-up protocol. The CON protocol was usual the preflight warm-up that

judo athletes perform during competition (Table 1). In the ULB condition, subjects performed half the volume of their usual precompetition warm-up followed by 2 sets of 5 repetitions of resistance band pull at maximal effort (Figure 2) and 2 sets of 5 repetitions of standing broad jump. For the LB condition, subjects performed half the volume of the usual precompetition warm-up followed by 3 sets of 5 repetitions of standing broad jump. For the standing broad jump, subjects were instructed to flex their

knees and hip to their individual preferred angle before push off and jump as far forward as possible. Subjects performed the 5 repetitions without a pause in between. A 1 minute recovery period was given between each set of resistance band pull and standing broad jump. The reduction in volume of the usual precompetition was an attempt to equalize the work done for each warm-up protocol. As exercises in each protocol differs, it was not possible to compare the total work done. However, the time taken to complete each warm-up protocol was similar. Subjects were asked to rate their perceive exertion upon completion of each warm-up protocol using the session RPE scale (8). The recovery period after the warm-up was set at 5 minutes before performing the HPT. Subsequently, subjects performed the SJFT after a 2 minutes recovery period. This recovery time

TABLE 2. Number of throws, heart rate and index from the SJFT ($n = 11$).

	CON	LB	ULB
SJFT Series A	4.9 ± 0.5	5.3 ± 0.6*	5.3 ± 0.5*
SJFT Series B	9.5 ± 1.1	9.5 ± 1.0	9.7 ± 0.8
SJFT Series C	9 ± 0.9	8.8 ± 0.9	9.3 ± 0.6
SJFT Total	23.4 ± 2.2	23.6 ± 2.3	24.3 ± 1.6†
HR _{post}	180 ± 10	180 ± 9	181 ± 8
HR _{post1min}	134 ± 16	140 ± 13	133 ± 9
Index	13.9 ± 1.7	13.6 ± 1.6	13.3 ± 0.9

*Indicates significant difference from CON ($p \leq 0.05$).

†Indicates significant difference from CON ($p < 0.01$).

period from PAP stimulus was selected as a meta-analysis of previous studies showed that 5–7 minutes was an optimal recovery period to maximize PAP effect (20). The other 2 testing sessions were separated by a 72 hours thereafter.

High Pull Test. This test was selected as it mimics the movement profile of a judo throw, whereby both upper and lower limbs are required and was previously used as a test to measure power output in an earlier study (23). The load for the HPT was set at half of each participant's body weight rounded off to the nearest 2.5 kg. The HPT was executed from a midhigh position where subjects adopt a knee flexion angle of 120–130°. Subjects were required to perform 3 repetitions of the exercise at maximal effort. Peak Power for each repetition was measured using a TENDO unit (Model V-104; TENDO Sports Machines, Trencin, Slovak Republic). The free end of the unit was attached to one end of the barbell with the unit was positioned linearly underneath the barbell. To calculate the power output in watts (W), the load being lifted was keyed into the analyzer. The TENDO unit computed the peak power according to the speed of the movement. The highest peak power attained was recorded.

Special Judo Fitness Test. This test was developed by Sterkowicz (22) and was reported to have a reliability value of 0.97. All subjects were grouped into groups of 3 based on their body mass (± 5 kg). Subjects remained in the same groups of 3 throughout the entire experimental period. During the SJFT, one participant (tori) was evaluated whereas 2 others (uke) received the throws. The tori was positioned in center with the 2 uke standing 3 m away from on each side. On signal, the tori ran to one uke and perform the ippon-seoi-nage throwing technique. He then ran toward the other uke and performed a throw on him. The tori had to complete as many throws as possible within the test time. The SJFT composed of 3 intervals (15, 30, and 30 seconds), with each interval separated by a 10 seconds recovery period. Performance was determined by the total number of throws completed. Heart rate (HR) was measured with HR monitor (RS400; Polar Electro Oy, Kempele, Finland) and recorded immediately (HR_{post}) and 1 minute after (HR_{post1min}) the test to calculate the following index which decreased with better test performance:

$$\text{Index} = (\text{HR}_{\text{post}} + \text{HR}_{\text{post1min}}) / \text{number of throws}$$

During the trial, the one tester was counting and recording the number of successful throws performed during each interval. The second tester kept track of the time and gave the verbal command to start and end each interval.

Statistical Analyses

All tested variables were expressed by Mean (1 *SD*). A repeated measure ANOVA was used to determine whether any differences in HPT and SJFT performance exist between all warm-up conditions. Significant differences among warm-up

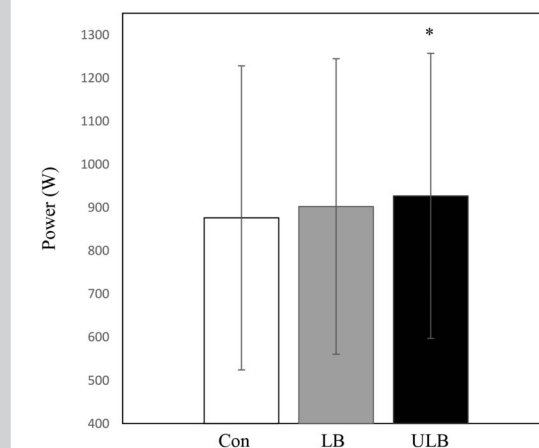


Figure 3. Power measurement from high pull test. *Indicates significant difference from CON ($p < 0.01$).

conditions were follow-up with pair wise comparisons with Bonferroni correction. Cohen's *d* effect sizes (ESs) were also calculated and reported with small ES being less than 0.4, moderate between 0.4 and 0.7, and large being greater than 0.7. All statistical significance was set at $p \leq 0.05$.

Pearson correlation analysis was selected to determine the association between peak power and SJFT performance. Correlational indices were set at: (a) small if $0 \leq |r| \leq 0.2$; (b) moderate if $0.2 < |r| \leq 0.5$; and (c) strong if $|r| > 0.5$.

RESULTS

The main results of the SJFT are presented in Table 2. The LB and ULB condition resulted in significant difference in

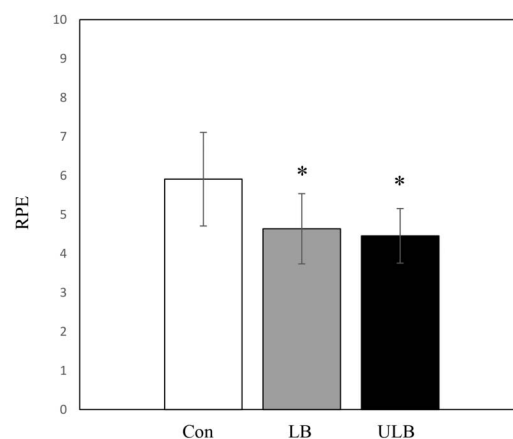


Figure 4. RPE ratings of the warm-up protocols. *Indicates significant difference from CON ($p < 0.01$).

number of throws in series A as compared to CON ($p \leq 0.05$). The ES for LB and ULB were 0.6 and 0.7, respectively, indicating a moderate effect. No significant differences were observed between conditions for series B and series C. The total number of throws performed was significantly higher than CON in ULB only ($p < 0.01$) with a moderate ES of 0.5. There were no significant differences in number of throws between LB and ULB. No significant differences in HR_{post} , $HR_{post1min}$ and index were found between conditions.

Power measurement during the HPT is presented in Figure 3. Only ULB resulted in a significantly higher power output as compared to CON ($p < 0.01$). However, ES analysis showed a small ES of 0.1. No difference was observed between LB and ULB. Pearson correlation test showed a positive correlation between peak power and the total number of throws performed during the SJFT ($r = 0.4$, $p \leq 0.05$).

Figure 4 presents the RPE values for the 3 warm-up protocols. Both LB and ULB resulted in significantly lower RPE values as compared to CON ($p < 0.01$). The ES for LB and ULB were 1.2 and 1.5 respectively, indicating a large effect. No difference was observed between LB and ULB.

DISCUSSION

The results of this study supported the first hypothesis as the LB and ULB protocols resulted in a higher number of throws in first series of SJFT as compared to a normal judo warm-up protocol. In addition, ULB resulted in higher number of total throws as compared to CON. The results also showed that power output was increased after performing the ULB protocol. Furthermore, it was shown that both LB and ULB resulted in lower RPE rating than CON. However, the second hypothesis was not supported as there were no differences in SJFT performance between LB and ULB.

The results of the current study was consistent with that of Miarka et al. (16) who also reported a better performance in the first series of SJFT after performing plyometric exercise as compared to the control condition. Most studies investigating the effect of PAP on exercise performance have mainly used heavy weight lifting as the PAP stimulus. To the author's knowledge, limited number of studies were conducted to directly compare the effects of a plyometric and resistance exercise as a PAP stimulus (1,6,7,16). The study by Da Silva Santos et al. (6) showed that total number of kicks performed by taekwondo athletes only increased after performing plyometric exercise in combination with heavy resistance exercise, whereas plyometric exercise and heavy resistance exercise alone showed no significant difference from control condition. In contrast, the study by de Villarreal et al. (7) showed that volleyball athletes improved in jumping performance after performing plyometric exercises but it did not show any significant difference as compared to heavy resistance exercise. Similarly, though the findings of Miarka et al. (16) showed that only plyometric exercise

resulted in better performance in the first series of the SJFT as compared to control condition, there was no difference between plyometric and heavy resistance exercise. However, Seitz and Haff (20) mentioned in their meta-analysis that plyometric might be more effective than a traditional resistance exercise in inducing potentiation as plyometric exercises are associated with preferential recruitment of type II motor units which is the mechanism underpinning PAP. Furthermore, to produce a PAP effect, the level of potentiation resulted from the exercise has to be greater than the resulted fatigue level. A plyometric exercise may cause less fatigue as compared to a traditional resistance exercise which can lead to a greater potentiation effect to be achieved. Another advantage of performing plyometric exercise over traditional resistance exercise is that athletes can choose to perform plyometric exercises that do not require equipment, for example, countermovement jump or standing broad jump. Athletes who intend to induce PAP effect might not have access to heavy weights at the competition venue. Therefore, performing plyometric exercise would be a better option.

This study included the ULB condition, as upper limb muscles are required during a judo throw action (10). The resistance band pull exercise was selected in an attempt to produce PAP effect as the movement of the exercise mimics the judo throw action. The results showed that performing the resistance band pull in combination with standing broad jump led to a significantly greater number of total throws during the SJFT and higher peak power during the HPT. In addition, correlational analysis showed that there was a moderate positive correlation between peak power and total number of throws. This suggests that being able to generate higher power output could be beneficial to judo performance. To date, there are very limited number of studies conducted to investigate the potentiating effects of variable resistance exercises. Among these was the study by Baker (2) which investigated on the effects of interspersing bench press with accommodating chains resistance between sets of bench throws. The results showed that peak power output increased by 4.8–7.7% after the intervention. In another study, Mina et al. (17) showed that replacing 35% of total weight lifted during a back squat with resistance from resistance band resulted in greater potentiating effect as compared to normal loaded squat. The author stated that including the variable resistance changed the loading pattern during the squat, which allowed for loading to be closer to maximal capacity throughout the range of movement. This could be the reason for the enhanced PAP effect. Although, the 2 aforementioned studies utilized variable resistance to induce PAP effect, the majority of the loads lifted during the exercises were still free weights. To the author's knowledge, apart from the current study, only Aandahl et al. (1) had investigated the potentiating effects of variable resistance alone. In this study, athletes with background of kickboxing and taekwondo performed kicks with elastic resistance attached to their legs during warm-up. Similar to the findings

of the current study, the results showed positive potentiating effect after kicking with elastic resistance as kicking velocity and quadriceps muscles activation was increased. The authors concluded that this was because of increased recruitment of higher order motor units, better synchronization of involved motor units and a reduced presynaptic inhibition.

Judo is a high intensity combat sport with a time structure of 20–30 seconds of activities and 10 seconds of inactivity during a 5 minutes bout (4). Therefore, it is advantageous for the athlete to recover as much as possible within the inactivity period (9). The index calculated from the SJFT could be an indication of the athlete's recovery rate. Miarka et al. (16) reported that a combination of plyometric and heavy resistance exercise before SJFT resulted in a lower index as compared to control condition. However, no differences were observed between conditions in the current study. This suggests that both LB and ULB have no effect on cardiovascular recovery.

As judo is an intensive sport, it would be logical to refrain from inducing high level of fatigue on the athlete during the warm-up process. The data from this study showed that both LB and ULB resulted in lower RPE values as compared to CON while still producing superior SJFT result. This suggests that it might not be necessary for judo athletes to perform high volume of warm-up with judo specific activities before competition if they include resistance band pull and standing broad jump into the warm-up protocol.

There were several limitations in this study. Firstly, there is no condition in this study whereby subjects did not perform any warm-up before performing HPT and SJFT. Therefore, the results of the study was not able to answer if the CON protocol was actually beneficial or detrimental to HPT and SJFT performance. Secondly, the resistance used for the resistance band pull was the highest resistance that subjects could pull through the full range of motion as there is currently no study to provide reference on the optimal resistance level to use when performing variable resistance exercise to induce PAP effect. Finally, 4 out of the 11 subjects in this study were below 18 years old. These athletes were selected for the study as they were part of the national training program and were efficient in performing the ippon-seio-nage throw technique. The athletes in the current study performed a total of 23.4–24.3 throws during the SJFT. This was comparable with the athletes in the study by Miarka et al. (16), where the athletes managed to perform a total of 23.7–24.3 throws. However, the physiological maturity of the athletes might have affected the results as it has been suggested that stronger athletes exhibit greater PAP effect (20). Future studies could investigate on the optimal resistance level to use when attempting to induce PAP effect and control for participant's age.

PRACTICAL APPLICATIONS

Judo is a high intensity sport that requires a high level of power when executing a throw technique. An ideal warm-up

protocol before each competition bout should be to minimize fatigue levels while maximizing the effect of muscular potentiation. The current study showed that judo athletes can increase their peak power and optimize their performance by including resistance band pull exercise and standing broad jump to their warm-up routine. These exercises can be performed in competition venues as they are low in logistical demand. In addition, coaches can also include these exercises into athlete's daily warm-up regime to increase power output during training. Finally, it is also possible for coaches to modify the movement resistance band pull exercise to mimic individual athlete's dominant throw technique as the resistance band is a flexible equipment.

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